AGENTIC AI CAPSTONE · FINAL REVIEW

Scheme Saathi

Finding the welfare schemes a household qualifies for.

Digvijay · Saurabh · Sharika · Snehal · Sathya

G4-Saathi · 15/11/2026

THE PROBLEM

Families miss schemes they qualify for

Hundreds of rules

Each scheme has its own age, income, residence and family conditions, written as prose on myScheme.

Edge cases decide

Age 59 against 60, income just over a limit, a pension already received: the answer turns on one fact.

Mistakes cost

A wrong “eligible” wastes a trip and documents; a wrong “not eligible” loses a benefit. Both erode trust.

Our answer: agents talk to the citizen in Hindi, Marathi or English, but reviewed rules in code decide eligibility, and a person approves every application.

ONE REQUEST, START TO FINISH

“My mother is 62, a widow, in Pune…”

PLACEHOLDER: redrawn architecture diagram

Numbered 1–7 to match the steps on the right, with the evaluation harness, claim judge and CI beside the request path.

1. **Input guardrails** block messages with Aadhaar, bank or phone numbers, and check for injection.
2. **Supervisor** routes and shows the citizen a plan (profile, schemes, rules, documents, draft), revised from the rule results.
3. **Intake** asks only the question that settles the most open verdicts.
4. **Eligibility analyst** shortlists schemes; the **rule engine** returns eligible, not eligible or can’t tell, with the clause.
5. **Document checker** lists what each scheme needs and flags mismatches.
6. **Application agent** drafts on the mock portal; the graph pauses for approval of the exact form.
7. **Output guardrails** block eligibility claims not from the rule engine and payment requests, and state unknowns plainly.

THE CORE DESIGN CHOICE

Code decides eligibility, not the model

Three answers

Eligible, not eligible, or can’t tell, with the missing facts named.

Fails closed

One unreviewed rule makes the scheme “can’t tell”. No guessing, no type coercion.

Cites the clause

Every verdict names the scheme clause it rests on, so a person can check it.

The rule engine is an MCP server. Agents may call it; they may not overrule it. An output guardrail blocks eligibility claims that did not come from it.

GOLD DATA

Everything is measured against checked answers

49

reviewed rule sets

Field, operator, value and cited clause, each checked against the myScheme text. 15 hand-coded; 34 drafted, then reviewed.

71

synthetic households

Built around rule boundaries; 58 English, 9 Marathi, 4 Hindi. Development: 21 cases, 39 verdicts. Held-out: 50 cases, 95 verdicts, 21 at a boundary.

Held-out cases run only with **--final**, and each final comparison was frozen before it ran.

SIGNATURE A · E3: DID THE LESSONS HELP?

Rules from prose, improved by Reflexion

An extractor turns scheme text into rules; a reflection step turns its errors on development schemes into lessons. A person approved 13 of 34 proposals; 12 were frozen before the held-out runs.

Condition F1	Without lessons	With frozen lessons
Held-out, 15 schemes, 3 runs each	0.880 (0.876–0.885)	0.868 (0.856–0.875)
Development, gpt-4o-mini	0.64	0.69–0.71
Development, gpt-4.1-mini	0.74–0.75	0.76

Finding: lessons helped where they were learned, not on new schemes. One lesson, L007, made M13 worse. A better model helped more than the lessons. Cost: about 37,500 tokens.

SIGNATURE B · E4: QUESTION ORDER

Ask the question that settles the most verdicts

Questions per conversation	Planned	Fixed form order	Full form
Development, candidate schemes	5.5	5.5	8.0
Held-out, candidate schemes	6.7	6.9	8.5
Development, all 49 schemes	147	163	163
Held-out, all 49 schemes	138	163	163

Finding: with a household's 1–4 candidate schemes, stopping early saves a fifth to a third of a full form and order hardly matters. With every scheme open, planning saves 10–15%, but 138 questions is still too many. **Choosing the right few schemes matters most.** Verdicts were 100% right in every setting.

E4 · THE SIGNATURE ABLATION

What if the model decided directly?

100%

Rule engine, 95 held-out verdicts

91%

gpt-4.1-mini from the scheme text, 3 runs (90.5–91.6%)

- **No boundary errors** either way: the model read “below” and “up to” correctly on all 21 boundary verdicts.
- **7 of its errors** are one scheme (C23): a reviewed pilot-state rule the text it saw does not state.
- **Its answers changed between runs**; the engine’s never do, cost nothing and cite the clause.

E1 · DOES THE AGENT TEAM JUSTIFY ITS COST?

A short conversation, but it misses schemes

Held-out, 50 cases	C0, given schemes	C1, given schemes	C0, all 49 open (E4)	C2 agent team
Knows the schemes?	yes	yes	no	no
Questions per case	6.7	none	138	9.3
Verdicts right	100%	100%	100%	54% (51–57%)
Cost per case	$0	$0.0004	$0	$0.0081
Median latency	0.13 s	6.3 s	no model	69 s

Finding: given the schemes, the fixed workflow wins. Without them it needs 138 questions; the agent team needs 9 but misses 39% of the right schemes. Its shortlist is the next fix; a later development run with eligibility text in the shortlist options got 29/39, against 22/39 (one run each).

Scheme Saathi · 9 · gpt-4o-mini for C1 and C2; citizen claude-haiku-5.5; C1, C2 × 3 runs · Source: evaluation report §2

E2 · DO THE DEFENCES WORK?

96 attacks, defences off and on

79% → 5%

Attack success, defences off → on (91 of 96 blocked)

0%

of 29 legitimate cases blocked with defences on

5

requests to misstate or hide facts still get through

- **Held:** blocked ID numbers, tool permissions, approval tied to the form, draft-rule gate, other households' data, IDs in traces, injected scheme text, and now a planted scam page and rules from unofficial sources.
- **Still open:** refusing requests to under-report income, overstate age or hide a fact.
- **Limit:** code defences only, and the attacks were visible while the defences were built.

HUMAN CONTROL · L14

Nothing is submitted without approval of the exact form

Pause

The graph stops with interrupt() and shows the form: approve, edit or reject.

Bound to the form

Approval carries the form's hash; the portal accepts only a signed, one-use token for that form.

Edit resets it

Any change makes a new form, so an earlier approval no longer works.

Only the application agent can reach the write server. Pre-tool checks enforce permissions and the tool-call budget; post-tool hooks log every call.

MEMORY · L9

Remember only what the family agrees to

- Long-term facts need consent, expire, and must be reconfirmed.
- Only allowlisted fields are kept; ID numbers never are.
- “Forget my data” deletes the household’s facts, consents and references.
- Each household’s memory is isolated; red-team attacks on it were blocked.

Memory on vs off

−16%

questions on a return visit with memory on (4.95 → 4.16 per visit, 150 visits), 0 verdicts changed

EVALUATION · L13

Can we trust the judge?

- Code checks first: verdicts, citations, forbidden actions.
- An LLM judge reads only what code cannot decide: does a reply make an eligibility claim? The harness refuses a judge from the agents' model family.
- Calibrated against hand labels with Cohen's kappa: a Claude judge checking GPT agents.

Calibration

κ 1.0

20/20 agreement with hand labels on 20 synthetic messages (claude-haiku-5.5 judge).

DEPLOYMENT AND OPERATIONS · L16

Running it, watching it

- FastAPI with streaming replies, behind the Streamlit app.
- Docker image of the app; Prometheus and Grafana run in Docker.
- Grafana dashboard: questions per chat, can't-tell rate, waiting approvals, masked-ID hits, cost per conversation.
- Alert when the can't-tell rate jumps: a sign of a corpus or extractor problem.
- CI runs the full test suite and builds the app and Grafana images.

Still to do

- One command that starts everything
- An evaluation gate that fails CI below a threshold
- One failed CI run, explained
- Install on a clean machine

LIMITS WE REPORT

What these numbers do not show

One labeller

The brief asks for two; there is no agreement score. Why: labelling 71 personas took the time we had.

E1 compares systems

The fixed workflow and single agent were given the candidate schemes; only the agent team chose its own. Why: they are upper bounds on accuracy.

Held-out used up

The sets are small, and the held-out cases were seen in later work; any later change needs a fresh test set. Why: fixes needed real failures to work from.

Code-level security tests

The red team tests code defences, with attacks known to the builders; a model-level test is still to do. Why: code tests are repeatable; model runs were limited by API credit.

Next: repeat and improve the agent team's shortlist, refuse requests to misstate facts, test the judge on real agent replies, then freeze and test on fresh cases.

LIVE DEMOS

Three things to see working

1

Widow pension in Hindi

From the son's message to an approved draft, with the clause behind each verdict.

2

Refusing to cheat

Refuses and explains when asked to under-report income; flags and ignores a scam page. [Scam page: blocked by the payment check. Misstatement refusal: pending]

3

"Forget my data"

The household's remembered facts are gone from the store.

Also ready to show: stop and resume a conversation, and an edit invalidating an earlier approval. A backup recording covers all of them. [Recording: pending]